MODULE 17 · SECTION 2 - GOOGLE CLOUD'S GEN AI OFFERINGS

Agent tools and the pre-trained APIs

The four tool types, the Google Cloud services agents reach, and the ready-made AI specialists you can call instead of training your own.

SECTION 2 of 4 EXAM WEIGHT ~35% MODULE 17 of 33

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An agent is only as useful as its tools. The exam asks **which tool, for which job**.

Four tool types an agent can use

TOOL TYPE	WHAT IT DOES
Extensions	The agent calls an API directly, itself
Functions	The model proposes a call; your app runs it
Data stores	Fresh facts from your own content
Plugins	Ready-made add-ons that bring a capability

REMEMBER

Think of a new hire's kit: a phone with numbers saved, a request form, a library card, and installed apps.

Extensions or functions?

EXTENSIONS

The agent makes the call

- Calls the API itself
- Runs on the agent's side
- Quick for standard services
- Less control over each call

FUNCTIONS

Your app makes the call

- Model returns a name and arguments
- Your application runs it
- You control access and checks
- Suits private or sensitive systems

THE TIEBREAKER

Your own code must control every call, for security or approval: functions. Vertex AI Extensions are deprecated, but the guide still tests the idea.

Data stores: the agent's library card

- **Fresh, specific facts** the model was never trained on.
- **Structured** content, such as product tables, and **unstructured**, such as PDFs.
- **Usually built** with Agent Search: this is RAG inside an agent.
- **Answers can cite** the document they came from.
- **Update the content,** and the agent knows straight away. No retraining.

> THE SIGNAL
>
> The agent must know facts it was not trained on, and they change often: a data store.

Google Cloud services agents reach

THE AGENT NEEDS	SERVICE
Live business records	Cloud SQL, AlloyDB, Spanner, Firestore
Analytics across the business	BigQuery
Files, images and recordings	Cloud Storage
A small piece of code on an event	Cloud Run functions
A whole service or API you host	Cloud Run

ON THE EXAM

A function call often lands on Cloud Run or Cloud Run functions, which then reads a database. The agent never touches the database directly.

Cloud Run functions or Cloud Run?

CLOUD RUN FUNCTIONS

One small job, on an event

- A single piece of code
- Runs when something happens
- Scales to zero; pay per use
- Formerly Cloud Functions

CLOUD RUN

A whole service

- A full app or API in a container
- Many endpoints, any language
- Runs as long as needed
- Also scales to zero

THE TIEBREAKER

"When a file arrives, do one thing": Cloud Run functions. "Host our API or app": Cloud Run. Older papers say Cloud Functions.

Pre-trained APIs are **ready-made specialists**. Hire one instead of training your own.

Speech and language

API	WHAT IT DOES
Speech-to-Text	Turns audio into text, in many languages
Text-to-Speech	Turns text into natural-sounding voices
Translation	Translates text across 100+ languages
Document Translation	Translates whole files, keeping the layout
Natural Language	Finds sentiment, entities and categories

THE TIEBREAKER

Watch the direction: speech to text is listening; text to speech is speaking.

Documents, images and video

API	WHAT IT DOES
Document AI	Pulls structured fields from documents
Cloud Vision	Labels, text, logos and objects in images
Video Intelligence	Labels, scenes, objects and text in video
API Library	The catalogue of every Google Cloud API

THE TIEBREAKER

Reading text in a photo: Vision. Turning a messy document into named fields: Document AI.

A pre-trained API, or prompt Gemini?

PRE-TRAINED API

A specialist for one job

- One well-defined task
- Structured, predictable output
- Built for high volume
- Often cheaper per item

GEMINI

A generalist that reasons

- Many tasks from one model
- Mixes text, images and audio
- Adapts with a prompt
- Wording varies run to run

THE TIEBREAKER

A standard, high-volume task with a dedicated API: the API. An unusual task that mixes inputs and needs reasoning: Gemini. They often work together.

Wrenfield's jobs, matched to tools

JOB	TOOL
52,000 invoices a month	Document AI, then a function posts them
48,000 Dutch descriptions	Translation, with a glossary
Care guides as Dutch PDFs	Document Translation
"Where is my order?"	A function that queries Cloud SQL
Policies that change weekly	A data store

THE TAKEAWAY

Each fix pairs an AI capability with a plain Google Cloud service. That pairing is what most real agents look like.

Phrases that point to each tool

THE QUESTION SAYS	CHOOSE
"Your own code must make the call"	Functions
"Answer from documents that change weekly"	A data store
"Extract fields from invoices and forms"	Document AI
"Translate the PDF, keep the layout"	Document Translation
"Detect logos or scenes in video"	Video Intelligence
"Run code when a file arrives"	Cloud Run functions

ON THE EXAM

Name the input and the output first. Most of these questions are decided by that pair.

> Wrenfield's returns agent must check order status in the private orders system.
>
> The security team insists that Wrenfield's own application makes every call, with its own credentials and checks.

Which tool type fits best?

A Extensions

B A data store

C Functions

D A plugin

◆ PAUSE AND DECIDE

Which tool type fits best?

A Extensions

The agent would call the API itself.

B A data store

Holds content for answers; it does not call systems.

C Functions

The model proposes the call; Wrenfield's app runs it.

D A plugin

A ready-made add-on, not a controlled private call.

THE SIGNAL

Your own code must make the call: functions.

> Wrenfield's staff agent must answer questions from 3,000 policy and product documents. The documents change every week.

Which tool type fits best?

A A function

B An extension

C Retrain the model every week

D A data store

◆ PAUSE AND DECIDE

Which tool type fits best?

A A function

Calls a system; it does not hold documents.

B An extension

Calls an API; it does not hold documents.

C Retrain the model every week

Slow, costly and always a week behind.

D A data store

Fresh facts from your content, updated without retraining.

THE SIGNAL

Facts the model never learned, changing often: a data store.

> About 52,000 supplier invoices and delivery notes arrive each month, as PDFs and scans in many layouts.
>
> Fourteen staff retype the supplier, order number and total by hand.

Which API should do the reading?

A Document AI

B Cloud Vision

C Translation

D Natural Language

◆ PAUSE AND DECIDE

Which API should do the reading?

A Document AI

Extracts named fields from documents in many layouts.

B Cloud Vision

Reads text in images, but not as named invoice fields.

C Translation

Translates; it does not extract fields.

D Natural Language

Finds sentiment and entities in prose, not invoice fields.

THE SIGNAL

Turn documents into named fields: Document AI.

> Wrenfield wants spoken announcements in every store, such as closing-time reminders. Staff will type the message, and it should play in a natural voice.

Which API fits best?

A Speech-to-Text

B Text-to-Speech

C Translation

D Video Intelligence

◆ PAUSE AND DECIDE

Which API fits best?

A Speech-to-Text

The wrong direction: audio into text.

B Text-to-Speech

Typed text out as a natural voice.

C Translation

Changes language, not text into sound.

D Video Intelligence

Analyses video; it does not speak.

THE SIGNAL

Text in, voice out: Text-to-Speech.

> For the Netherlands launch, Wrenfield's 12 supplier care guides must be in Dutch.
>
> They are PDFs full of tables, diagrams and careful layout, which must stay intact.

Which option fits best?

A Copy the text out and use the Translation API

B Paste each guide into the Gemini app

C Document Translation

D Cloud Vision

◆ PAUSE AND DECIDE

Which option fits best?

A Copy the text out and use the Translation API

Translates the words, but the layout is lost.

B Paste each guide into the Gemini app

Manual, inconsistent, and the layout is lost.

C Document Translation

Translates whole files and keeps the layout.

D Cloud Vision

Reads images; it does not translate.

THE SIGNAL

Translate the file and keep the layout: Document Translation.

> Wrenfield has 200,000 written product reviews.
>
> The product team wants each review's sentiment, and the products it mentions, as structured data, cheaply and at scale.

Which API fits best?

A Google Vids

B Video Intelligence

C Speech-to-Text

D Natural Language

◆ PAUSE AND DECIDE

Which API fits best?

A Google Vids

Makes videos; it does not analyse text.

B Video Intelligence

Analyses video, not written reviews.

C Speech-to-Text

Transcribes audio; these reviews are already text.

D Natural Language

Sentiment and entities, as structured output, at scale.

THE SIGNAL

Sentiment and entities in text, at scale: Natural Language.

> Marketing has 900 customer-submitted videos for a competition.
>
> Each must be tagged automatically for the objects shown, any visible logos, and its scene changes.

Which API fits best?

A Cloud Vision

B Video Intelligence

C Speech-to-Text

D Document AI

◆ PAUSE AND DECIDE

Which API fits best?

A Cloud Vision

Analyses still images, not video over time.

B Video Intelligence

Objects, logos and scene changes across video.

C Speech-to-Text

Transcribes speech, not what is on screen.

D Document AI

Reads documents, not video.

THE SIGNAL

Objects, logos and scenes in video: Video Intelligence.

> Each time a new invoice PDF lands in Cloud Storage, a small piece of code should send it to Document AI.
>
> The team wants nothing running, and nothing to pay, between invoices.

Where should that code run?

A Cloud Run functions

B Cloud Run, as a full service

C BigQuery

D The Gemini app

◆ PAUSE AND DECIDE

Where should that code run?

A Cloud Run functions

One small job, triggered by the file, scaling to zero.

B Cloud Run, as a full service

Works, but a whole service for one small job.

C BigQuery

Analyses data; it does not run event code.

D The Gemini app

A chat app, not a place to run code.

THE SIGNAL

Run one small piece of code when a file arrives: Cloud Run functions.

> A customer posts a handwritten letter and a photo of a damaged wardrobe.
>
> Wrenfield wants an agent to read both, work out what went wrong and who is at fault, and draft a fair reply.

Which capability fits best?

A Cloud Vision, to read the handwriting

B Document AI form parsing

C Gemini, a multimodal model, on Agent Platform

D The Natural Language API

◆ PAUSE AND DECIDE

Which capability fits best?

A Cloud Vision, to read the handwriting

Reads the words, but cannot judge or reply.

B Document AI form parsing

Extracts form fields; this is a free-form letter.

C Gemini, a multimodal model, on Agent Platform

Reads both inputs, reasons about them, and writes.

D The Natural Language API

Finds sentiment, but cannot see the photo or decide.

THE SIGNAL

Mixed inputs plus reasoning: Gemini.

> An exam question describes a tool that lets an agent call an external API itself, on the agent's side.
>
> It is set up with examples that teach the agent when to use each API.

Which tool type is described?

A A data store

B A plugin

C A function

D An extension

◆ PAUSE AND DECIDE

Which tool type is described?

A A data store

Holds content; it does not call APIs.

B A plugin

A packaged add-on, not this pattern.

C A function

Your app runs the call, not the agent.

D An extension

The agent calls the API directly, guided by examples.

THE SIGNAL

The agent makes the call itself: an extension.

Module 17 in five lines

- Four tool types: extensions, functions, data stores and plugins.
- Functions let your own app make the call; data stores bring fresh facts.
- Databases hold records, Cloud Storage holds files; Cloud Run functions react to events.
- Pre-trained APIs are ready-made specialists for speech, language, documents, images, video.
- Standard, high-volume task: the API. Mixed inputs and reasoning: Gemini.